Zeyu XIE

32301184@stu.hzcu.edu.cn | mylovesong.github.io

EDUCATION

Hangzhou City University (HZCU) **09/2023-07/2027 (Expected)**

Bachelor s' Degree in Computer Science and Technology, Top Innovation Class **GPA:** 3.61/4.0; **Average:** 82.7/100

Research oriented honours cohort with approximately 10 students per year.

Modules include: Integrative Computer Science Project Design (95), Reading and Writing Scientific Papers (91), Introduction to Artificial Intelligence (86), Linear Algebra (85).

Zhejiang University (ZJU) **09/2025-07/2026**

Visiting Student in Computer Science and Technology **GPA:** 3.84/4.3; **Average:** 86.38/100

Modules include: Computer Vision (92), Computer Animation (89), Fundamentals of Software Engineering (89), Computational Photography (88).

RESEARCH INTERESTS

Multimodal Foundation Models; Vision Language Action Models; Embodied AI; Open World Generalization and Adaptive Learning.

PUBLICATIONS

Ming, Z., **Xie, Z.**, Zhang, C., Su, K., Yuan, C., & Chua, T. S. (2025). Robust visual food recognition for enriching nutrition knowledge bases. *IEEE Transactions on Multimedia*, *27*, 2825–2835. https://doi.org/10.1109/tmm.2025.3542962

- **First student author**; led the experiments and manuscript writing under my advisor's supervision.
- Achieved the highest food recognition accuracy across 10% to 45% label noise settings; at 30% noise, the proposed cAUM and AccGap framework reached 84.49% accuracy, outperforming the strongest baseline by 9.07 percentage points.

RESEARCH EXPERIENCE

Research on Modelling, Evolution and Intervention of Individualised Blood Glucose Response Mechanisms in Free-Choice Dietary Scenarios **04/2026-04/2027 (Expected)**

National Undergraduate Innovation Project | Project Lead

- Developed a population-informed personal-history module that estimates individual nutrient-response patterns from prior meals and pre-meal glucose, generating 15 meal-specific features for 90-minute glucose prediction with frozen TabPFN weights.
- Outperformed the SOTA TabPFN-3.5 baseline, reducing MAE, TWED, and observed-interval iAUC error by **0.87%**, **0.70%**, and **2.05%**, respectively, across five random seeds on a 7,297-meal development set.
- Validated the contributions of population priors and correctly paired meal-response histories through controlled ablations against simpler personal summaries and shuffled historical pairings.

Research on Lightweight Target Detection and Tracking System for High-Altitude Drones Based on Multimodal Fusion **04/2025-10/2026 (Expected)**

National Undergraduate Innovation Project | Core Member

- Proposed feature-fusion ideas for multimodal small-object detection and discussed lightweight model designs with the team.

Research on Food Image and Nutrition Multimodal Representation and Visual Recognition Methods for Diabetes Dietary Records **04/2024-04/2026**

Provincial Undergraduate Innovation Project | Project Lead

- Led food selection and synonym expansion using LLM and suppressed noisy labels with cAUM/AccGap, which resulted in the IEEE TMM paper.
- Selected FoodnKB entries, collected web images and conducted multimodal representation learning.

Acute Leukemia Prognosis Modeling from Cell and Karyotype Microscopy 2023–2024

National Undergraduate Innovation Project | Team Member

- Project focus: microscopy image segmentation with limited labels and interpretable prognosis modeling.

IndusMind: E-Commerce Product Search and Query–Product Relevance System **2025-2026**
Project Lead

- Built a hybrid product-search pipeline combining field-weighted BM25, BGE-M3 dense retrieval, reciprocal-rank fusion, and Qwen3 reranking, improving nDCG@10 from 0.706 to 0.778 on 384 held-out queries from the Wayfair ANnotation Dataset (WANDS), a public e-commerce search benchmark.
- Developed a query–product relevance module and integrated the retrieval and ranking pipeline into search, matching, and human-review APIs for a catalog of 164,900 products.

"Nutritional Transparency" Campus Dietary Health Information Disclosure Programme **2024**
Project Lead

- Led a team of 40 to research campus catering and establish a dietary nutrition knowledge base to develop recommendation algorithms.
- Served over 12,000 staff and students by launching a dietary recommendation website that provided public access to nutritional information and personalised dietary recommendations.

PATENTS & SOFTWARE COPYRIGHTS

Method, system, electronic device, and storage medium for generating hypoglycaemia warning signals based on a calibration-aware decision layer [Under review] **04/2026**
First Inventor Application No. 202610501761.8

Nutritionist Intelligent Assistance Backend Management System **01/2025**
Core Developer Certificate No. 14690887

Multimodal Intelligent Nutritionist Mini Programme **01/2025**
Core Developer Certificate No. 14690975

WORK EXPERIENCE

Shaoxing Ruoming Import & Export Co., Ltd. **07/2026-10/2026 (Expected)**
Research and Development Intern

- Adapted the IndusMind retrieval and relevance pipeline to real product catalogues, supporting image, keyword, and natural-language product matching.
- Built a lightweight enterprise RAG system and compared keyword-based with semantic retrieval on real product queries.
- Reduced automatic product-matching errors by approximately 40% on an internally labelled evaluation set using confidence thresholds and human review.

AWARDS & HONOURS

Government Scholarship, Zhejiang Provincial **2026**
First-Class Scholarship, for Academic Competitions **2025–2026**
Third-Class Scholarship, for Academic Excellence **2025–2026**
First Prize, Higher Education Group, Provincial Contest, Network Track, 10th Huawei ICT Competition, China Practice Competition. **2026**
First-Class Scholarship, Science & Technology Innovation (Individual). **2024-2025**
Second Prize, University Mathematical Modelling Competition. **2024-2025**
Second-Class Scholarship for Outstanding Academic Research. **2024-2025**
Third-Class Scholarship for Academic Excellence. **2024-2025**
International Exchange Scholarship (Funded). **2024-2025**

EXTRACURRICULAR ACTIVITIES

Centre for Continuing Education, University of Macau **07/2025**
Visiting Student

- Completed the training course on New Quality Productive Forces - AI and Interdisciplinary Study.

SKILLS

- **Programming Skills:** Python, PyTorch, MATLAB, Node.js, Express.

- **Machine Learning Skills:** Multimodal modelling, robust machine learning, noisy label learning, model evaluation.
- **Foundation Model Skills:** LoRA, Quantisation, RAG, FAISS, LangChain, Qwen models.
- **Engineering Development Skills:** BM25, BGE-M3, FAISS, reciprocal-rank fusion, neural reranking, API development.
- **Languages:** Mandarin Chinese (native); English (IELTS overall: 6.5).